

Experiment No: 9

Title: Implement localStorage and sessionStorage to save user preferences like theme settings.

Software/Tools Required:

JavaScript, HTML, CSS and Code Editor.

Theory:

Web Storage: localStorage and sessionStorage in JavaScript

Modern web applications often need to store user data locally in the browser to improve user experience and maintain preferences between page visits. JavaScript provides the **Web Storage API**, which includes **localStorage** and **sessionStorage**. These storage mechanisms allow websites to save data as key-value pairs directly in the user's browser without requiring a database or server-side storage.

1. Web Storage API

The Web Storage API provides a simple way to store data in the browser. Unlike cookies, web storage can store larger amounts of data and does not send the stored information to the server with every request.

The two types of web storage are:

- localStorage
- sessionStorage

2. localStorage

localStorage stores data permanently in the browser until it is explicitly removed by the user or application. The stored data remains available even after the browser is closed and reopened.

Example:

```
localStorage.setItem("theme", "dark");
```

To retrieve stored data:

```
let theme = localStorage.getItem("theme");
```

To remove stored data:

```
localStorage.removeItem("theme");
```

3. sessionStorage

sessionStorage stores data only for the duration of the current browser session. The data is automatically deleted when the browser tab or window is closed.

Example:

```
sessionStorage.setItem("username", "John");
```

To retrieve stored data:

```
let user = sessionStorage.getItem("username");
```

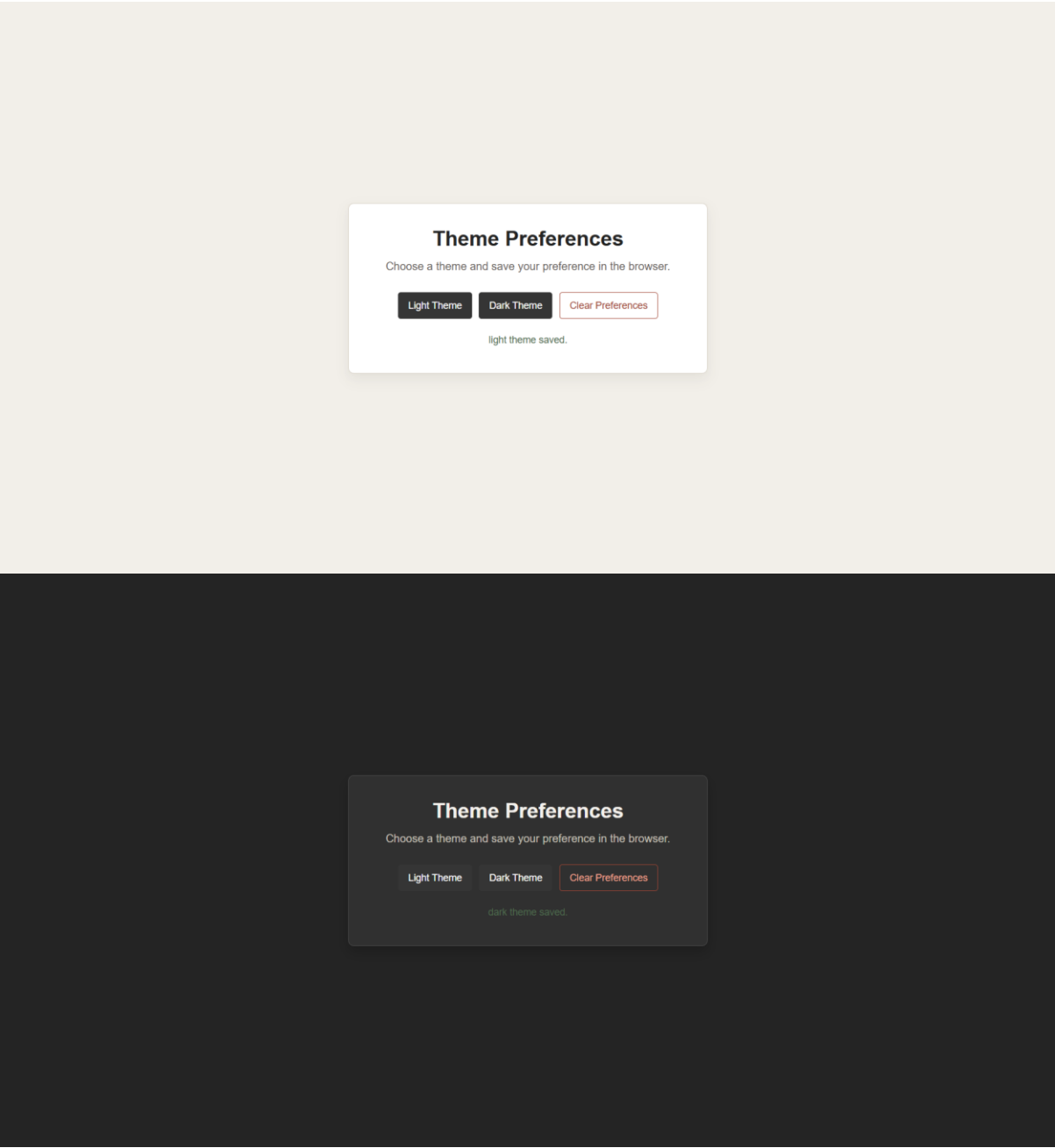
To remove stored data:

```
sessionStorage.removeItem("username");
```

Case Study:

A Schedule Planner can use `localStorage` to save the selected schedule details such as day and time, so the information remains available even after refreshing the webpage. `sessionStorage` can be used to store the selected topic, number of schedules viewed, and recent topics during the current session. This helps the application remember the user's schedule preferences and provide a simple personalized experience.

Output



CaseStudy

XML SEMINAR

Seminar Schedule Planner

Click any time or topic to view its schedule details.

Day	Schedule		Topic
	Begin	End	
Monday	8:00 a.m.	5:00 p.m.	Introduction to XML
			Validity: DTD and Relax NG
Tuesday	8:00 a.m.	11:00 a.m.	XPath
	11:00 a.m.	2:00 p.m.	XSL Transformations
	2:00 p.m.	5:00 p.m.	XSL Transformations
Wednesday	8:00 a.m.	12:00 p.m.	XSL Formatting Objects

☐ Begin time ☐ End time

Last Selected Schedule

No schedule selected yet.

Schedules Viewed This Session: 0
Recent Views: None

Change Topic

Enter a new topic

Update Topic

Clear History

XML SEMINAR

Seminar Schedule Planner

Click any time or topic to view its schedule details.

Day	Schedule		Topic
	Begin	End	
Monday	8:00 a.m.	5:00 p.m.	Database
			Database
Tuesday	8:00 a.m.	11:00 a.m.	XPath
	11:00 a.m.	2:00 p.m.	XSL Transformations
	2:00 p.m.	5:00 p.m.	XSL Transformations
Wednesday	8:00 a.m.	12:00 p.m.	XSL Formatting Objects

☐ Begin time ☐ End time

Last Selected Schedule

Day: Tuesday | Topic: XSL Transformations | Time: 2:00 p.m. - 5:00 p.m.

Schedules Viewed This Session: 4
Recent Views: XSL Transformations, XSL Transformations, XPath

Change Topic

XSL Transformations

Update Topic

Clear History

Conclusion:

The Web Storage API provides efficient client-side storage through **localStorage** and **sessionStorage**. While `localStorage` retains data across browser sessions, `sessionStorage` stores data only for the current session. These storage mechanisms are widely used to save user preferences such as theme settings, enabling websites to provide a personalized and convenient user experience without relying on server-side storage.

